Name: _____

TOPIC TEST 2

# Chapter 8 Continuous random samples and the normal distribution

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Eight multiple-choice questions (8 marks).
- Part B: Ten short-answer questions (32 marks).
- Part C: Two analysis questions (17 marks).

## Part A

### Multiple-choice questions

Eight multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

**1** Which of the following could be probability density functions on the intervals given?

I $f(x) = 0.4$ on the interval $[10, 14]$

II $f(x) = \dfrac{x - 3}{5}$ on the interval $[3, 8]$

III $f(x) = \begin{cases} 0.16(x - 2) \text{ for } 2 \leq x \leq 4.5 \\ -0.16(x - 7) \text{ for } 4.5 \leq x \leq 7 \end{cases}$

IV $f(x) = 2x - 12$ on the interval $[6, 7]$

**A** All of I–IV

**B** I and II only

**C** III and IV only

**D** Only IV

**E** II and III only

**[1 mark]**

**2** The probability density function of the continuous random variable $X$ on the interval $[1, 4]$ is $\text{pdf}(x) = \frac{6x - 5 - x^2}{9}$. What is the cumulative probability function?

**A** $\frac{-x(x^2 - 9x + 15)}{27}$

**B** $\frac{x(6x - 5 - x^2)}{9}$

**C** $\frac{3x^2 - 5x - \frac{1}{3}x^3}{9x}$

**D** $\frac{9x - 15 - x^2}{27}$

**E** $\frac{2x^2 - 5x - \frac{1}{3}x}{9}$

**[1 mark]**

**3** The expected value of a uniform probability density function on the interval $[10, 48]$ is:

**A** 24

**B** 29

**C** 19

**D** $\frac{1}{38}$

**E** $\frac{5}{24}$

**[1 mark]**

**4** The variance of a uniform probability density function on the interval $[5, 25]$ is closest to:

**A** 15

**B** 5.8

**C** 4

**D** 33.3

**E** 30

**[1 mark]**

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**5** What is the approximate area under the standard normal distribution for $0.3 \leq Z \leq 1.8$?

**A** 0.15

**B** 0.35

**C** 0.067

**D** 0.58

**E** 0.43

**[1 mark]**

**6** What is the standard normal score for $x = 14$ for a random normal variable with mean 22 and standard deviation 6?

**A** 0.64

**B** $-0.64$

**C** 1.33

**D** $-1.33$

**E** $-0.75$

**[1 mark]**

**7** $X$ is a random normal variable with a mean of 42.5 and a standard deviation of 6.8. Find the probability that a random value of $X$ lies between 39.2 and 46.

**A** 0.03988

**B** 0.04475

**C** 0.4364

**D** 0.05215

**E** 0.3829

**[1 mark]**

**8** What is the value of the quantile $t_{0.9}$ for the normal random variable $X$ with mean 12.8 and standard deviation 4.35?

**A** 0.3843

**B** 16.715

**C** 1.2816

**D** 20.7539

**E** 18.3747

**[1 mark]**

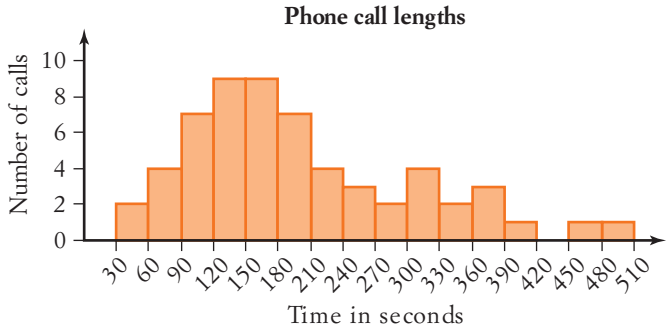
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# Part B

## Short-answer questions

Ten short-answer questions
32 marks
Show your working where appropriate.

**1** The histogram below shows the lengths of phone calls to the sales office of a surfboard manufacturer. Use the histogram to find the probability that the next phone call is at least 2 minutes long.




**[3 marks]**

**2** Make a probability density function using $8x - x^2 - 7$ on the interval $[1, 7]$.

**[3 marks]**

**3** **a** Find the probability density function for the cumulative probability function $\text{cdf}(X = x) = \dfrac{x^2(6 - x)}{32}$ defined on the interval $[0, 4]$.

**b** Find the probability $P(0.9 \leq X < 1.9)$.

**[4 marks]**

**4** **a** Construct a uniform probability density function $f(x = X)$ for the interval $[6, 20]$.

**b** Find the probability $P(7 \leq X \leq 10 \text{ or } 12 < X < 16)$

**[3 marks]**

**5** The distance between two points is given as 45 m, to the nearest metre. Construct a triangular probability function for the distance and find its mean and standard deviation.

**[5 marks]**

**6** A continuous random variable $X$ is transformed to the random variable $Y$ according to the equation $Y = 2X - 3$. The mean and standard deviation of $Y$ are 18.9 and 4.8 respectively. What are the mean and standard deviation of $X$?

**[3 marks]**

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**7** What is the probability density function of a normal distribution with $\mu = 30$ and $\sigma = 6$?

**[2 marks]**

**8** Hayley got 65 for Physics, where the class average was 55 and the standard deviation was 8. For Chemistry, she got only 60, but for Chemistry the class average was 46 and the standard deviation was 9. In which test did she do better, assuming the classes were of equal ability?

**[5 marks]**

**9** $Y$ is a standard normal variable. Find each of the following:

**a** $P(Y < 0.36)$

**b** $P(-1.2 \leq Y \leq 0.7)$

**[2 marks]**

**10** The average mass of Year 12 boys is 79 kg, with a standard deviation of 8.5 kg. What is the first quartile?

**[2 marks]**

# Part C

## Analysis questions

Two analysis questions
17 marks
Show your working where appropriate.

**1** The javelins thrown on a school sports day can land any distance between 10 m and 35 m from the line. The most common distance is about 20 m.

**a** Construct a triangular probability density function to model the javelin distances.

**b** Find the probability of a randomly selected student throwing 28 m, to the nearest metre.

**c** Find the mean and standard deviation of the javelin distances.

**[9 marks]**

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**2** 30% of the students doing a calculus problem finish it in 9 minutes, while 70% finish it in 12 minutes. Assuming a normal distribution, how long would it take for 90% to finish the problem?

**[8 marks]**

**[Total marks: 57]**

## Answers

### Part A – Multiple-choice questions [1 mark each]

**1** Only III and IV have integrals of 1 $\therefore$ **C**

**2** $\int \frac{6x - 5 - x^2}{9} dx$

$= \frac{3x^2 - 5x - \frac{1}{3}x^3}{9}$

$= \frac{-x(x^2 - 9x + 15)}{27}$

$\therefore$ **A**

**3** $\frac{10 + 48}{2} = 29$

$\therefore$ **B**

**4** $\frac{(25 - 5)^2}{12}$ 33.3

$\therefore$ **D**

**5** Using calculator, $P(0.3 \leq Z \leq 1.8) = 0.3461\ldots$

$\therefore$ **B**

**6** $Z = \frac{14 - 22}{6} \approx -1.33$

$\therefore$ **D**

**7** Using calculator, $P(39.2 \leq X \leq 46) \approx 0.3829$

$\therefore$ **E**

**8** Using calculator invNorm(0.9, 12.8, 4.35) $\approx$ 18.3747

$\therefore$ **E**

### Part B – Short-answer questions

**1** Total number of calls = 59 **[1 mark]**

Number at least 2 minutes long = 46 **[1 mark]**

Probability of call at least 2 minutes long = $\frac{46}{59} \approx 0.78$ **[1 mark]**

**2** $\int_1^7 (8x - x^2 - 7)dx = \left[4x^2 - \frac{x^3}{3} - 7x\right]_1^7 = 36$ **[2 marks]**

pdf is $f(X = x) = \frac{1}{36}(8x - x^2 - 7)$ **[1 mark]**

**3 a** $f(x) = \dfrac{d}{dx}\left(\dfrac{6x^2 - x^3}{32}\right)$ **[2 marks]**

$= \dfrac{12x - 3x^2}{32}$

**b** $P(0.9 \leq X < 1.9) = \text{cdf}(1.9) - \text{cdf}(0.9) \approx 0.3334$ **[2 marks]**

**4 a** $f(x) = \dfrac{1}{14}$ **[1 mark]**

**b** $P(7 \leq X \leq 10 \text{ or } 12 < X < 16) = \displaystyle\int_7^{10} \frac{1}{14}dx + \int_{12}^{16} \frac{1}{14}dx = 0.5$ **[2 marks]**

**5** The function will be 1 wide (i.e. maximum error of 0.5), so for the area to be 1, the height in the centre will be of height 2. **[1 mark for height]**

Making each section a straight line gives $f(x) = \begin{cases} \dfrac{1}{4}(x - 44.5) & \text{for } 44.5 \leq x \leq 45 \\ -\dfrac{1}{4}(x - 45.5) & \text{for } 45 \leq x \leq 45.5 \end{cases}$

**[1 mark for slopes, 1 for each part of the equation]**

mean = 45 (by symmetry) **[1 mark]**

$\text{SD} = \dfrac{45.5 - 44.5}{2\sqrt{6}} = \dfrac{1}{2\sqrt{6}} \quad 0.2041$ **[1 mark]**

**6** $X = \dfrac{Y + 3}{2}$, so $E(X) = \dfrac{18.9 + 3}{2} = 10.95$, $\text{SD}(Y) = \frac{1}{2} \times 4.8 = 2.4$ **[3 marks]**

**7** $p(x) = \dfrac{1}{6\sqrt{2\pi}} e^{-\frac{(x-30)^2}{2\times 6^2}} = \dfrac{1}{6\sqrt{2\pi}} e^{-\frac{(x-30)^2}{72}}$ **[2 marks]**

**8** $Z_{\text{Physics}} = \dfrac{65 - 55}{8} = 1\dfrac{1}{4}$ **[2 marks]**

$Z_{\text{Chemistry}} = \dfrac{60 - 46}{9} = 1\dfrac{5}{9} > 1\dfrac{1}{4}$ **[2 marks]**

Francis did better on the Chemistry test. **[1 mark]**

**9** Using calculator:

**a** $P(Y < 0.36) \approx 0.6406$ **[1 mark]**

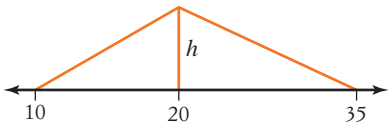
**b** $P(-1.2 \leq Y \leq 0.7) \approx 0.6430$ **[1 mark]**

**10** Using the inverse normal function with area = 0.25 on a calculator, Q3 ≈ 73.3 kg. **[2 marks]**

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**Part C – Analysis questions**

**1 a** The distribution is not symmetrical, so the areas either side of the maximum are not equal.




**[1 mark]**

Let the height at 20 be $h$.

Then $\frac{1}{2} \times 10 \times h + \frac{1}{2} \times 15 \times h = 1$

Thus $h = \frac{1}{12.5} = 0.08$. **[1 mark]**

For $10 \leq x \leq 20, f(x) = \frac{0.08}{10}(x - 10) = 0.008(x - 10)$ **[1 mark]**

For $20 \leq x \leq 35\ f(x) = \frac{-0.08}{15}(x - 35) = -\frac{2}{375}(x - 35) \approx -0.00533(x - 35)$ **[1 mark]**

**b** $\int_{27.5}^{28.5} -\frac{2}{375}(x - 35)dx = 0.037\overline{3}$

The probability of a randomly selected student throwing 28 m is about 0.0373.

$$\text{Mean} = \int_{10}^{35} f(x)x\,dx = \int_{10}^{20} (0.008(x - 10))x\,dx + \int_{20}^{35} \left(-\frac{2}{375}(x - 35)\right)x\,dx = 21.66\ldots$$

**[2 marks for correct integral, 2 for correct evaluation]**

The mean javelin throw is about 21.7 m. **[1 mark]**

**2** Since it is a normal distribution, $m = 10.5$ by symmetry. **[1 mark]**

For a standard normal distribution, the 0.3 quantile is at $Z = -0.5244\ldots$ from the inverse normal distribution function. **[2 marks]**

Thus $-0.5244\ldots = \frac{9 - 10.5}{\sigma}$, so $\sigma = 2.8604\ldots$ **[2 marks]**

Using the inverse normal function, invnorm(0.9,10.5, 2.8604…) = 14.165… **[2 marks]**

So 90% will have completed the problem in about 14.2 minutes. **[1 mark]**

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